

PARTS LIST

Semiconductors:

IC1	- ATtiny13 microcontroller
IC2	- 7805, 5V regulator
T1	- BC547 npn transistor
D1	- 1N4001 rectifier diode

Resistors (all $\frac{1}{4}$ -watt, ± 5 per cent carbon):

R1, R2	- 10-kilo-ohm
R3	- 2-kilo-ohm

Capacitors:

C1	- 0.1 μ F ceramic disk
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Miscellaneous:

RL1	- 12V, 1C/O relay
B1	- 12V bulb
S1	- Tactile switch
	- RBS04 tilt switch

fixing the problem, the bulb turns off automatically. Relay RL1 is powered by a 12V battery. IC 7805 provides regulated 5V supply to the microcontroller.

Construction

An actual-size, single-side PCB for car-bonnet light control using tilt sensor is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any error. Use an IC base for the microcontroller. Before inserting the IC, check the supply voltage.

As shown in Fig. 1, fix the tilt sensor on the inner side of the car bonnet with m-seal (to protect it from high temperature) such that when the bonnet is lifted, the tilt switch closes to switch on the light. Bulb B1 is fixed inside the bonnet away from the engine such that it lights the engine properly

Overview of Bulb Control Operation

Bonnet condition	Tilt sensor contact	Status of PB1	Status of PB0	Relay	Bulb light
Open	Closed	Low	High	Energised	On
Closed	Open	High	Low	De-energised	Off

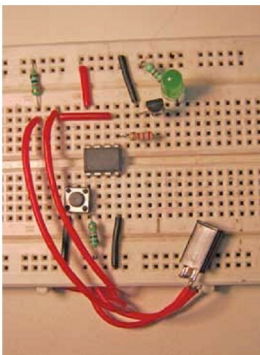


Fig. 5: Car bonnet light control using tilt sensor wired on a breadboard

when the bonnet is lifted. Connectors are provided on the PCB to extend the tilt sensor and bulb.

When lifting of the bonnet is detected by the microcontroller, its pin 5 is set to drive transistor T1 into conduction. Relay RL1 energises to switch bulb B1 'on'. The operation of the circuit is summarised in the table.

Software

The source program is written in Assembly language and assembled using AVR Studio to generate the Intel hex code. The code is well commented and easy to understand.

The generated hex code is programmed into the chip using a suitable programmer. The device is shipped with CKSEL="10", SUT="10" and CKDIV8 programmed. The default clock source setting is therefore the internal R-C oscillator running at 9.6 MHz with longest startup time and an initial system clock prescaling of 8. This default setting ensures that all users can make the desired clock source setting during programming.

The ATtiny13 is programmed to sense close/open contact of the tilt sensor and energise/de-energise the relay contact, which, in turn, switches the bulb 'on'/'off'.

The ATtiny13 works in power-down mode when it is idle, drawing current in micro-amperes. This results in basically no power consumption by the circuit, except when the relay is energised.

EFY Note. The source code of this article is available on www.efymag.com website.

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TILT.ASM

```
.include "TINI3DEF.INC"
.org 0x00
rjmp start
.org INT0addr
rjmp myint0
.org 0x020
myint0: IN R19, SREG
PUSH R19 ;stores sreg on stack
nop
IN XL,PINB ; read portb pins
NOP
SBRX XL,PB1; skip next instruction
if pb1 is logic high
SBI PORTB,PB1; switch on the relay
SBRX XL,PB1 ;skip next instruction
if pb1 is logic low
CBI PORTB,PB0 ;else switch off the
relay
```

```
NEXT: POP R19
OUT SREG,R19
reti ;return from interrupt and
set I-Bit in SREG
;Program Starts here
.org 0x050
start: LDI R19,LOW(RAMEND)
OUT SP,R19 ;setup stack
sbi ACSR,ACD ;shut down comparator
to save power
LDI R16,(0<<PB0)|(1<<PB1)
out portb,r16 ;switch off
the relay and
;enable pull-up on input
ldi r16,(1<<DDB0)|(0<<DDB1)
out ddrb,r16;configures PORTB0 as
output, PB1 as input
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```
ldi r21,(1<<SE)|(1<<SM0)|(1<<ISC00);
any change on INTO
; and power down mode
enabled
out mcucr,r21 ;
ldi r23,(1<<INT0)
out gims,r23 ;enable the
INT0
LDI R20,(1<<SE);enable sleep
OUT MCUCR,R20
sei ;global INT set
REPEAT: nop
sleep
nop
rjmp REPEAT
.exit
```